Tidal disruption of stars by black holes of 10^6–10^8 solar masses in nearby galaxies

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Stars in galactic nuclei can be captured or tidally disrupted by a central black hole. Some debris would be ejected at high speed; the remainder would be swallowed by the hole, causing a bright flare lasting at most a few years. Such phenomena are compatible with the presence of 10^6–10^8 $M_\odot$ holes in the nuclei of many nearby galaxies. Stellar disruption may have interesting consequences in our own Galactic Centre if a $\sim 10^6$ $M_\odot$ hole lurks there.

TIDAL capture and disruption of stars was explored in the 1970s as a possible way of fuelling active galactic nuclei (AGNs). The outcome of these studies was that tidal disruption could not supply a quasar-level fuelling rate unless the stellar concentrations around the hole were so great that other processes (such as stellar collisions) would supervene and release still more gas[1-4]. The process has now, however, become interesting in an obverse context. The dynamics of stars in the inner regions of nearby galaxies such as M31 and M32 indicate the presence of central concentrated dark masses, and one would like to know whether these might indeed be massive black holes[5-9]. As the number of surrounding stars, and their motions, are roughly known, the capture rate by the putative central hole can be estimated; the question then arises of whether the apparent quiescence of the nuclei of these galaxies is compatible with a massive hole's presence. The answer depends on what happens to the debris from each disrupted star. How much is accreted or expelled? What is the associated luminosity or radiative efficiency? And how long does it take to digest or expel the debris from one star, in comparison with the interval between successive captures?

The conjecture that there might be a massive black hole in our own Galactic Centre goes back more than 15 years (see, for example, ref. 10). There is some dynamical evidence for a mass concentration within the central 0.1 pc (ref. 11); the peculiarities of the unresolved radio source[12] indicate some unique object right at the centre, even if much of the activity and structure outside the central parsec can be accounted for in conventional terms. But the situation here is still somewhat ambiguous: in a recent review, Genzel and Townes[13] rate the evidence for a black hole of order 10^6 $M_\odot$ as 'substantial but not fully convincing'.

Studies of the quasar and AGN population at redshifts $z > 2$, lead to the well known inference that dead remnants—black holes lurking in the centres of relatively quiescent galaxies—should be widespread. The lifetime of AGNs is uncertain: there could be one generation, with typical individual lifetimes $\gtrsim 2 \times 10^9$ years; alternatively, many successive generations could trace out relatively brief life cycles over the period when the entire population evolves. In the former case, quasars should leave rare and hugely massive remnants; in the latter, remnants would be more common, but less individually massive. In the latter case (but not the former), Seyfert galaxies could be interpreted as reactivated quasars. Evidence on the masses, and frequency of occurrence, of black holes in present-day galaxies, therefore offers important clues to quasar lifetimes, and to the relationships between different classes of AGNs; it may also indicate how many galactic mergers, leading to coalescence of black holes, have occurred since the epoch $z = 2$.

The best diagnostic for a black hole's presence would be some inevitable concomitant that cannot be explained in any other way. A candidate may be the distinctive manifestations of tidal capture and disruption of stars. If the stellar density in the central few parsecs is known, it is in principle straightforward to estimate how often a star gets close enough to the central hole to be tidally disrupted (or at least captured). The debris from one solar-mass star per ten thousand years, swallowed steadily with 10 per cent radiative efficiency, would yield a luminosity of 6×10^{41} erg s^{-1}—higher than is observed from M31 or M32, or indeed from our own Galactic Centre. Some authors, particularly Ozernoi and his collaborators[14-16], believe such considerations may actually rule out a 10^6 $M_\odot$ hole at our Galactic Centre.

Several features of the stellar capture process have as yet not been fully analysed, and create uncertainties in this line of argument: (1) What fraction of the debris goes down the hole, rather than being expelled? (2) What is the radiative efficiency for the accretion process? In other words, how many ergs are radiated for each gram that is swallowed? (3) How long does it take to 'digest' or expel the debris from one star? In particular, how does the flare duration and decay timescale for such a process compare with the interval between one stellar disruption and the next? Were this time much shorter than the mean interval between successive captures, one might expect bright flares with short duty cycles; observational constraints would not then be stringent until we had observed enough candidate holes to constitute a proper 'ensemble average'.

This paper is concerned with such phenomena, with particular reference to the nearest galaxies, where the masses of the holes (if they are indeed present) are perhaps of order 10^7 $M_\odot$. I consider the fate of stars that approach the hole closely enough to get captured, showing that some debris would be expelled at $\gtrsim 10^4$ km s^{-1}; the remainder accretes on to the hole, yielding high luminosities and perhaps a further phase of radiation-driven outflow. Some implications of putative black holes in our Galactic Centre and in other nearby galaxies are then described.

Gravitational influence on surrounding stars

There have been several analyses of how a central mass influences the orbits and spatial distribution of stars in the inner regions of its host galaxy[17-20]. A central mass M_h would influence the dynamics at all radii where $(GM_h/r)^{1/2} \gtrsim \sigma(r)$, the stellar velocity dispersion at r. If the stellar distribution in the galaxy had a well defined core, characterized by a radius r_C and a stellar density N_* (the virial velocity of the stars then being $\sigma \simeq (Gm_*/r_C)^{1/2}$), and the central hole had a mass $M_h \ll N_* r_C^3$, then a typical star in the core would spend most of its time moving under the combined influence of the other stars rather than the hole; the hole would dominate the dynamics only within a radius $r_h = GM_h/\sigma^2 = (M_h/N_* r_C^3) r_C$. In the case of M31, for instance

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(refs 7-9), there is no well defined r_C—there is nonetheless evidence for a characteristic radius r_h within which σ rises (as does the rotational velocity) and the inferred mass-to-light ratio becomes anomalously large for a purely stellar population.

A star interacting with a massive hole cannot be treated as a 'point mass' if its gets so close to the hole that it becomes vulnerable to tidal distortions and even disruption. Such effects become important when the pericentre r_{min} becomes as small as the 'tidal radius'

$$r_T \simeq 5\times10^{12}\, M_6^{1/3}\, (r_*/r_\odot)\, (m_*/m_\odot)^{-1/3}\ \text{cm} \qquad (1)$$

where M_6 denotes the hole's mass in units of $10^6\, M_\odot$. Essentially, r_T (which is $\ll r_h$) is the distance from the hole at which M_h/r^3 equals the mean internal density of the passing star. (r_T is, in order of magnitude, the same as the Roche radius: the latter, though a precisely defined quantity, is only strictly applicable to a star in a circular orbit with synchronized spin, which is not the relevant situation here.) The escape velocity from the half-mass radius of a solar-type star is $\sim10^3$ km s^{-1}, and its specific gravitational self-binding energy is $\sim10^{-5}$ c^2. Slightly less close flybys with pericentre two or three times larger than r_T would be sufficient to dislodge the outer layers.

Exactly how frequently a star enters this 'zone of vulnerability' depends on the stellar velocity distribution. In a simple case when the velocities are isotropic at r_h, the frequency with which a solar type star would pass within a distance $r_{min}(\ll r_h)$ of the hole is

$$\approx 10^{-4}\, M_6^{4/3}\left(\frac{N_*}{10^5\ \text{pc}^{-3}}\right)\left(\frac{\sigma}{100\ \text{km s}^{-1}}\right)^{-1}\left(\frac{r_{min}}{r_T}\right)\ \text{yr}^{-1} \qquad (2)$$

This 'fiducial' capture rate could be inaccurate for a number of reasons. The actual rate could be lower than that given by equation (2), even if the initial distribution were isotropic, if the near-radial 'loss cone' orbits were depleted faster than they could be replenished[18,19]. Replenishment could be slow if it occurred on the two-body star-star diffusion timescale, rather than by some collective instability or in a triaxial potential. If the relaxation timescale were short enough to ensure replenishment of the loss cone, then the same diffusion processes could build up a 'cusp' in the stellar distribution near the hole[17,18]; and a population bound to the hole could arise in other ways[20,21]. Capture or disruption of these stars could then augment the rate given by equation (2). If such a cusp were constituted by ordinary solar type stars, stellar collisions would overwhelm 'Coulomb' encounters within a radius GM_h/v_*^2, where $v_* = (GM_*/r_*)^{1/2}$. For holes of $<10^8\, M_\odot$, however, none of these complications is likely to seriously modify the simple estimate (2).

For hole masses $>10^7\, M_\odot$, complete disruption of solar-type stars occurs sufficiently close to the hole for a newtonian approximation to be inadequate. (Relativistic precession effects on orbiting debris are, moreover, important even for lower M_h, as mentioned later.)

The probability of a pericentric distance within r_{min} ($\ll r_h$) is proportional to the first power of r_{min} (see equation (2)). Complete disruption requires $r_{min} < r_T$. For r_{min} somewhat larger than r_T, a star could lose its outer layers, but the denser interior would survive: this intermediate possibility is specially important for giants. For a range of still larger values of r_{min} (but still, at least for main sequence stars, only exceeding r_T a few times), a solar-type star loses no material, but gets so distorted during the flyby that the subsequent internal dissipation (when it readjusts) might exceed its incoming kinetic energy $1/2\ m_*\sigma^2$; energy conservation then implies that the star would thereafter be on an orbit bound to the hole[22-24].

Specific binding energy of debris elements

The observational consequences of stellar capture and disruption depend on what happens to the debris. Consider first a solar-type star that passes somewhat closer to the hole than r_T and is disrupted in a single flyby. The energy required to tear the star apart (that is, the star's self-binding energy) is supplied at the expense of the orbital kinetic energy (which at r_T is larger by $\sim(M/m_*)^{2/3}$). Unless there were some explosive energy input, which would be expected only of the star passed several times closer than r_T (see ref. 25), on average the debris would be bound to the hole, unless the star was initially on a hyperbolic orbit with asymptotic velocity $> v_*$, which is 1,000 km s^{-1} for solar-type stars.

Several effects would, however, impart orbital binding energies spread widely about this mean to gas from different parts of the disrupting star; this spread crucially influences what we would actually observe. The dominant effect is the following. While falling inwards towards the hole, the star would develop a quadrupole distortion which attains an amplitude of order unity by the time of disruption at $r = r_T$. The resultant gravitational torque would 'spin it up' to a good fraction of its co-rotation angular velocity by the time it gets disrupted: it would consequently, by that stage, be spinning at close to its break-up angular velocity. The parts on the 'outside track' furthest from the hole would therefore have an extra velocity, over and above the orbital velocity $v_{orb} \simeq (2GM/r_T)^{1/2} \simeq c(r_g/r_T)^{1/2}$, of order $v_* = (m_*/M_h)^{1/3} v_{orb}$; those closest to the hole would have a comparable velocity deficit. Moreover, the slower-moving gas on the 'inside track' is deeper in the potential well by an amount $\sim(GM_h/r_T)(r_*/r_T) \simeq (Gm_*/r_*)$. There would consequently be a spread of order $v_{orb}\Delta v$, where $\Delta v \simeq v_*$, in the energies of different bits of debris. Other processes during the flyby—for instance impulses from shocks during the drastic compression and distortion of the stellar material—could further enhance Δv.

Even though the mean specific binding energy of the debris to the hole would be positive, and comparable with the self-binding energy (Gm_*/r_*) of the original star, the spread about this mean is larger by $(M_h/m_*)^{1/3}$—a factor which is >100 for hole masses M_h in the range relevant to galactic nuclei. Whenever a solar-type star passed within the tidal disruption radius r_T, some of the debris would be flung out on hyperbolic orbits with escape velocities up to $\sim10^4$ km s^{-1}. (This important qualitative point was apparently first appreciated by Lacy *et al.*[26], and discussed by Rees[27] in the context of our Galactic Centre). When the central hole has a mass $M_6 > 1$, the bound debris would be on orbits with characteristic specific binding energy $\gtrsim 10^{-3}$ c^2 rather than $\sim10^{-5}$ c^2. The actual mass-fractions in a particular energy range, and the precise high-and low-energy cut-offs, would depend on the details; when a specific assumption is needed for illustrative purposes I shall assume that the mass-fractions are uniformly distributed in energy throughout the range from E_{min} (bound to hole) to E_{max} (unbound), where $E_{min} = -(Gm_*/r_*)\,((M_h/m_*)^{1/3}+1)$ and $E_{max} = (Gm_*/r_*)\,((M_h/m_*)^{1/3}-1)$.

Timescale for swallowing bound debris

The average specific binding energy of debris bound to the hole (just over half the total mass, if we adopt the illustrative model for the energy distribution mentioned above, and assume $M_h \gg m_*$) is $\sim\frac{1}{2}(Gm_*/r_*)\,(M_h/m_*)^{1/3}$. The bound orbits are very eccentric; for solar-type stars the orbital major axis of even the most tightly-bound debris is $\sim10^3\, M_6^{-1/3}\, r_g$, and the period is only $t_i \simeq 0.03\, M_6^{1/2}$ yr: note that this is shorter by a factor $\sim10^3\, M_6^{1/2}$ than the orbital period if the specific binding energy were (Gm_*/r_*), as had been assumed in pioneering discussions of this topic (such as refs, 2, 28). For $M_6 < 100$, this is certainly small compared with the interval between successive disruptions (equation (2)). Unless it takes many orbital periods to swallow the bound gas, the debris from each star would be digested separately—in contrast to the debris cloud model of quasars[2], where the disruptions are more frequent but the orbital periods of the debris ($\propto M_h^{1/2}$) are longer. I shall now give a reason why the gas could not persist for more than very few orbits without being swallowed or expelled.

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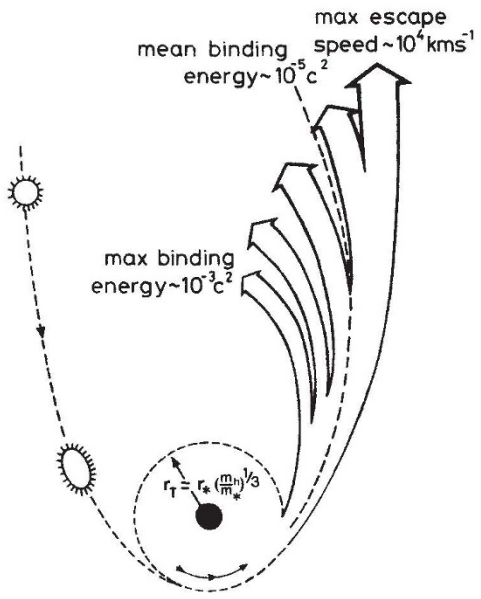


Fig. 1 A solar-type star approaching a massive black hole on a parabolic orbit with pericentre distance r_T is distorted and spun up during infall, and then tidally disrupted. The average specific binding energy of the debris to the hole is $\sim 10^{-5}\,c^2$ (of order the self-binding energy of the original star). However, the *spread* in this energy [of order $v\Delta v$, where $v = c(r_T/r_g)^{-1/2}$ and $\Delta v \simeq (Gm_*/r_*)^{1/2}$] is $\sim 10^{-3}\,c^2$ for hole masses $M_h \gtrsim 10^6\,M_\odot$. Almost half the debris would therefore escape on hyperbolic orbits with speeds up to $\sim 10^4\,km\,s^{-1}$; the most tightly bound debris would traverse an elliptical orbit with major axis $\sim 10^3\,r_g$ before returning to $r \simeq r_T$.

The range of orbital periods is large: if there were an equal mass-fraction in each interval of binding energy, then half the bound debris would have periods $>2^{3/2}$ longer than the innermost debris and a (small) mass fraction f would have periods longer by $>f^{-3/2}$. If the gaseous debris suffered no internal dissipation due to high viscosity or shocks, it would, after one or two orbital periods, form a highly elliptical disc with a big spread in apocentric distances between the most and least bound orbits, but where at pericentre the orbits were all squeezed in a range $(\Delta r/r) \simeq 0.01\,r_T$ for $M_6 \simeq 1$ (compare ref. 16). But we can readily see that a low-viscosity thin disc hypothesis is not tenable.

At each pericentre passage, where $r = r_T$, the relativistic orbital precession amounts to $\sim 3(r_g/r_T)$ radians, which for solar-type stars is $\sim 0.2\,M_6^{2/3}$ radians. This means that internal shocks with velocities of order $c(r_g/r_T)^{1/2}$ are inevitable once the most tightly bound gas has traced out more than one orbit. The debris raining down from 10^3–$10^4\,r_g$, all with specific angular momentum $\sim c r_T^{1/2} r_g^{1/2}$ appropriate to circular orbits at $r = r_T$, would, after little more than its free-fall time, settle into an axisymmetric torus, supported by radiation pressure, whose density maximum occurred in a ring with radius $\sim r_T$. This torus starts to form when the most tightly bound debris falls back in—at a time t_i after the disruptive flyby—and is thereafter fed by the injection of infalling matter at a rate that drops off as $t^{-5/2}$ for $t \gg t_i$ (again adopting the assumption of equal mass fractions in equal energy intervals). When a solar-type star has been disrupted by a close passage with pericentre distance $\sim r_T$, this infall rate is roughly given by

$$\dot{M} \simeq 25\,M_6^{-1/2}(t/t_i)^{-5/2} \quad \text{solar masses per year} \qquad (3)$$

The orbital period at r_T is a few hours ($\sim 10^{-3}$ yr). The viscous dissipation time for a thick torus would be α^{-1} orbital periods, α being the usual viscosity parameter; α would have to be below 10^{-3} for the dissipation time to exceed 1 yr. Such a low viscosity may be precluded by dynamical instability; moreover, the hole's spin axis would generally be misaligned with the star's orbital angular momentum, the resultant Lense-Thirring precession of the torus then being an extra cause of viscosity.

We then come up against another limit. The binding energy associated with a mass m_* in circular orbit at $r = r_T$ cannot be radiated away on a shorter timescale than $2.5 \times 10^8\,(m_*/M_h)(r_T/r_g)^{-1}$ yr without exceeding the hole's Eddington luminosity. This sets a minimum timescale of $\sim 5\,M_6^{-1/3}$ years for a solar mass of material to deflate into a thin ring at $r = r_T$, even if there were no further viscous dissipation, and $50\,\varepsilon_{0.1} M_6^{-1}$ yr for it to be swallowed by the hole with radiative efficiency $\varepsilon = 0.1\,\varepsilon_{0.1}$. Equivalently, the hole cannot accept matter, with radiative efficiency ε, at a rate exceeding

$$0.02\,\varepsilon_{0.1}^{-1} M_6 \quad \text{solar masses per year} \qquad (4)$$

The radiative efficiency depends on the viscous timescales in the inner axisymmetric flow pattern at $r < r_T$.

From equations (3) and (4) we readily infer that, if (as seems likely) the viscosity were high enough to process all the material within the infall timescale, then the luminosity L could not remain as high as the Eddington luminosity for longer than

$$17\,\varepsilon_{0.1}^{-0.4} M_6^{-0.6}\ t_i \sim 0.5\,M_6^{-0.1}\,\varepsilon_{0.1}^{0.4} \quad \text{yr} \qquad (5)$$

L would thereafter fall off as $t^{-5/2}$—indeed, the fall-off would probably be even steeper eventually , because the radiative efficiency of thermal processes in accretion flows generally falls off for very low $\dot{M}$; when $t \gg t_i$ the prime radiative output may be low-level non-thermal emission.

When the mass supply (3) exceeds the swallowing rate that (4) permits for high efficiency, there are two possibilities: (i) the material may all be swallowed, ε adjusting downwards so that all can be accepted without $\varepsilon \dot{M} c^2$ exceeding the Eddington luminosity. This could come about, in a manner familiar from studies of steady accretion tori, if a flow pattern established itself around the hole whose innermost parts followed orbits with binding energy close to zero. Material could then spill into the hole from such orbits with corresponding low radiative efficiency. (ii) Only a small fraction of the injected material may be swallowed, yielding radiation with high efficiency, the surplus being expelled in a radiation-driven wind.

One cannot (except by detailed considerations of the spectrum of the emergent radiation and so on) distinguish between these two possibilities. They do, however, differ importantly with regard to the rate at which the hole gains mass for a given stellar disruption rate (2), and the likelihood of a high-speed radiation-driven outflow.

A flare could persist steadily with $L \simeq L_E$ for longer than (5) only if the viscosity were so implausibly low that the bulk of the mass could be stored for many years in a 'reservoir' at $r = r_T$ (where the dynamical timescale is a few hours). In all other circumstances, the bulk of the debris would be swallowed or expelled rapidly compared with the interval between successive stellar captures—the only conspicuous luminosity being a flare (predominantly of thermal ultraviolet or X-ray emission) with peak intensity $L = L_E$, fading within a few years. The integrated output from this flare could, in principle, amount to a few per cent of the star's rest mass, but would probably be far less, because most debris would be 'fed' to the hole at $t = t_i$ far more rapidly than it could be accepted if ε were high; much would then escape in a radiatively-driven directed outflow or be swallowed inefficiently.

Unbound debris

When a star passes close enough to be disrupted in a single flyby, almost half the debris escapes on hyperbolic orbits. On the assumption (used above in connection with the bound orbits) that there is an equal mass-fraction in each energy interval up to E_{max}, the kinetic energy of the ejecta is $\sim 10^{51} M_6^{2/3}$ ergs. The material would be concentrated in a cone or 'fan' close to the

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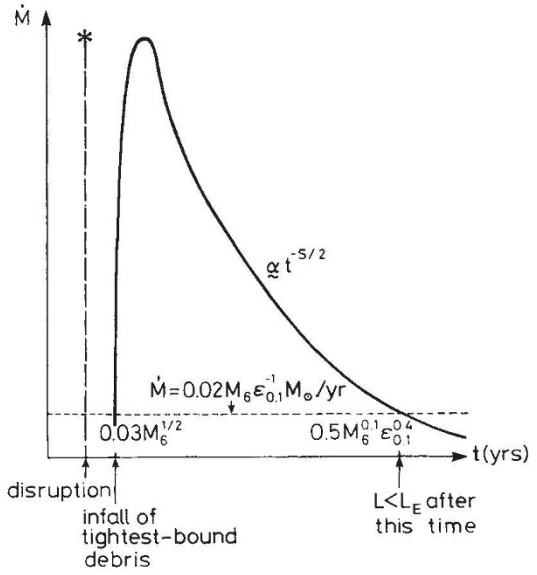


Fig. 2 When a disruptive flyby occurs with pericentric distance r_T, the bound debris starts to rain down on to the hole after a timelag $\sim 0.03\ (M_h/10^6\ M_\odot)^{1/2}$ years. This diagram shows schematically the subsequent behaviour of $\dot{M}$. The peak infall rate is highly 'super-Eddington'. This debris forms a torus at $r = r_T$ within which radiation pressure is dominant. Its subsequent flow is controlled by viscosity, which is likely to operate on a timescale only a modest multiple of the rotation period at $\sim r_T$ (only a few hours). A thermal luminosity $\sim L_E$ can only be maintained for ≤ 1 year; thereafter the 'flare' would rapidly fade.

orbital plane. Adiabatic cooling would severely reduce the internal radiative content before the debris became translucent. The 'prompt' radiation from the unbound debris would therefore release less than the initial energy content of the star (just as a supernova would be optically inconspicuous were it not for continuing energy injection in the months after the explosion). There would therefore be no conspicuous flare until, as discussed above, the bound debris fell back onto the hole after a time delay t_i. The main observable effects of the outflowing material would occur when it was decelerated by running into external diffuse matter.

Ultraclose passages and relativistic effects

The behaviour of stars passing well within the tidal radius r_T exhibits special features, some of which have been discussed in a series of paper by Carter, Luminet and collaborators[25,29-31]. Such stars are not only elongated along the orbital direction, but are even more severely compressed into a prolate shape (that is, a pancake aligned in the orbital plane). This compresson is halted by a shock, raising the matter (which then rebounds perpendicular to the orbital plane) to a higher adiabat. Also, there is the possibility of explosive energy release: the p-p reaction is too slow to release much energy on a dynamical timescale; however, proton capture on C, N and O can yield (for solar abundances) enough energy to unbind the star. The mean specific binding energy of the debris to the hole may then become negative, rather than being $\sim(Gm_*/r_*)$. But we have seen that, even in the latter case, about half the debris is unbound because of the spread $v_{orb}\Delta v$ in the energies at pericentre (where $\Delta v \simeq v_*$). If a nuclear explosion were to enhance Δv by some further factor, then the mass fraction escaping on hyperbolic orbits would still be ~ 0.5, but the range of energies of the debris would be increased: some would escape with higher hyperbolic velocities, whereas some would be even more tightly bound than before. To quantify this, one would need to know the angular distribution (in a frame sharing the star's mean motion) of the post-shock velocities, because velocity boosts perpendicular to the orbital plane yield only a second-order contribution, and are therefore less important than those in the plane.

The orbits for which extreme compression occurs would enter regions where relativistic effects were more important than for those with $r_{min} = r_T$. The orbits are then not ellipses, but may turn through 2π or even more. A test particle not exactly in the orbital plane (which in the case of an elliptical orbit would cross the plane just once at pericentre), may then have two or more traversals. There is then the possibility of multiple shocks[30,31].

For hole masses $M_h > 10^8\ M_\odot$, solar-type stars cannot be disrupted without entering the strongly relativistic domain. The form of the black hole (Schwarzschild or Kerr?) then has an important quantitative effect, as does (for a rotating Kerr hole) the orientation of the stellar orbit relative to the hole spin axis. Stars on counter-rotating orbits are more readily captured with the result that Kerr holes would spin down if they gained mass primarily from stellar capture[32]. When the hole mass is $\gg 10^8\ M_\odot$, most main sequence stars would be swallowed whole ($r_T < r_g$), and only giants would generate debris outside the hole[1,2].

Tidal dissipation without complete disruption?

A star which does not pass close enough to the hole to suffer complete disruption may nevertheless be tidally distorted; if the associated dissipation exceeds $m_*\sigma^2$, the star may then be captured in an elliptical orbit around the hole, and suffer further dissipation on each successive pericentric passage. This process has been primarily discussed in the context of globular clusters[22-24], where the star is of comparable mass to the compact object (stellar mass black hole or neutron star?); after capture, the orbit degrades and circularizes at a radius a few times r_T. One's first thought might therefore be that, in the present context, solar-type stars could be captured into near-circular orbits very tightly bound to a central massive hole. Such orbits at $r = r_T$ would have periods of a few hours (independent of M_h), and the orbital velocity would be $\sim 0.2\ M_6^{1/3}\ c$—vastly higher, of course, than the speeds involved in ordinary binary systems.

But there is a crucial new aspect to this tidal-capture mechanism when a very massive black hole is involved. In acquiring a tightly bound circular orbit of radius r_T, the star must dispose of an amount of energy $m_*c^2(r_T/r_g)^{-1}$—larger by a factor $(M_h/m_*)^{2/3}$ than its entire original gravitational self-binding energy Gm_*^2/r_*. The circularization would therefore need to occur much slower than the star's thermal timescale. Of course, the star would, after the first few passages, acquire a spin rate synchronized with its orbital angular velocity at pericentre, and thereafter the dissipation could be reduced somewhat below the rates calculated in refs 22-24.

Until appropriately detailed three-dimensional gas-dynamical calculations can be done, the fate of such stars will be an open question. It would seem more likely that, after the orbit had been 'ground down' by the first few passages (and the frequency of subsequent pericentre passages had become correspondingly higher), the star would inflate and get fully disrupted. The gaseous debris would then evolve essentially as the bound fraction of the debris from the tidally disrupted stars discussed earlier. More uncertain would be the fate of a giant star, where the dissipation, occurring primarily in the envelope, could leave the core relatively undistorted and unscathed. (If internal dissipation in the resultant torus were exceedingly low, the debris could deflate on the timescale $\sim 5\ M_6^{-1/3}$ years into a thin ring at $r = r_T$ which could in principle be vulnerable to gravitational instability, leading to rebirth of a star (compare ref. 33). But this would require an effective viscosity parameter $< 10^{-4}$, which is perhaps implausibly low.)

There is one further interesting mechanism for injecting stars into small tightly-bound orbits (though not necessarily as small as $r = r_T$. Diffusive stellar-dynamical processes could not inject stars into orbits with specific binding energies exceeding v_*^2—at such velocities disruptive stellar collisions dominate 'Coulomb' encounters. But a fraction of the stars in the core of the galaxy

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might well be binaries: tidal disruption of a binary by the hole, equivalent to a three-body exchange process, could (for reasons analogous to those already discussed in the context of the stellar debris) eject one member of the original binary on a high-speed hyperbolic orbit, leaving the other in an orbit tightly bound to the hole. Three-body calculations of this process have recently been presented by Hills[21]. The pericentre for such stars would be of order the 'tidal radius' for the original binary, and therefore well outside the tidal-disruption radius r_T for an individual star. Subsequent tidal circularization would therefore not occur fast enough to trigger disruption, and some slow frictional process (such as drag on gas) could then perhaps grind down the orbit until it approached $r = r_T$.

Despite the uncertainties (and perhaps improbabilities), the processes mentioned above indicate that it is far from inconceivable for a solar-type star to become established in an orbit near $r = r_T$. Its gravitational radiation decay time would then be

$$t_{GR} \simeq 10^6 M_6^{-2/3} \text{ yr} \quad (6)$$

typically long compared with the interval (equation (2)) between successive captures. Drag due to interaction with gas may degrade the orbit rather faster, but any star that did manage to get into a tightly bound orbit near $r = r_T$ could remain there until another star got captured on to (or settled into) a similar orbit. Any such star might be hard to detect directly, even in our own Galactic Centre (though an ordinary star with an anomalously high orbital velocity would offer gratifyingly unambiguous evidence for a massive black hole there). In an active galactic nucleus, such a star might modulate the intensity of any radiation due to gas accretion or electromagnetic effects. The search for strictly periodic variability in continuum emission from galactic nuclei, though a long shot, offers sufficient payoff to merit some continuing effort. Such stellar orbits woud display interesting relativistic effects. For instance, if the central hole were spinning, and the star's orbital angular momentum were misaligned with respect to the hole's spin axis, then Lense-Thirring precession of the orbital plane would have a timescale of $(J/J_{max})^{-1}\ (r_g/c)\ (r_T/r_g)^3$—only of the order of months for parameters relevant to the Galactic Centre, if the hole's angular momentum J were comparable with J_{max}, the maximum attainable by a Kerr metric.

A black hole at the Galactic Centre?

The dynamical evidence for a massive black hole at our Galactic Centre (reviewed in ref. 13) is based primarily on interpretation of gas motions within the central parsec. These data, together with the unresolved radio source[12], increasingly suggest that there is some unique object right at the centre, even if much of the activity and structure outside the central parsec can be accounted for in conventional terms. The situation is, however, confused by the fact that the brightest part of the central infrared source IRS 16 appears displaced from the radio source[34].

Because of intervening obscuration, we do not directly know the velocity dispersion of stars within the central parsec of our Galaxy; observations of the gas do, however, tell us something about the total mass in that region. Bailey[35] showed that, if the entire central mass concentration were a star cluster, then at a radius 0.1 pc the stars would be so closely packed that runaway dynamical evolution would take $<10^9$ yr. Unless we are observing at a privileged epoch in the history of our galactic nucleus, this supports the idea that a massive central object may already have formed.

The black hole hypothesis receives at least circumstantial support from the new evidence for black holes in nearby galaxies[5-9]. Perhaps all spiral galaxies pass intermittently through Seyfert phases: the black hole could then exist as a relic of earlier Seyfert-style outbursts, being reactivated when the fuelling rate was boosted. Recent evidence that the compact central radio source has a transverse motion <40 km s^{-1} (ref. 36) suggest that this source, whatever it is, indeed lurks right at our Galaxy's dynamical centre. If a massive hole were present in the central regions of any galaxy, then dynamical friction would settle it quickly into the true dynamical centre.

The small central radio component has an unusual (rising) spectrum; if there were a central hole with $M_6 = 1$, then the radio emission could come from plasma participating in a low-level accretion flow and gravitationally bound to the hole[33]: tenuous plasma would be unable to cool on the inflow timescale, the only substantial radiative output coming from within 1,000 Schwarzschild radii, where even the average electron could be relativistic with energy $\gtrsim 1$ Mev. There are, admittedly, alternative models that do not require a massive hole[37-39]. On the other hand, a massive hole accreting at a low rate would almost inevitably yield radio emission: the absence of a peculiar radio source at the Galactic Centre would therefore have been evidence against such an object there.

It would be a coincidence (at more than the 1% level) if we were now observing the Galactic Centre just at the time of a flare or stellar disruption. There is therefore no problem with the hypothesis that a star is disrupted once every $\sim 10^4$ yr, part being swallowed and part ejected—it is certainly acceptable that the mean luminosity from the centre should exceed what we now observe. Special interest attaches to ths particular galactic nucleus not only because it is 'close to home' but because it may exhibit residual traces of individual stellar-disruption events. One speculative possibility is the following.

The volume within ~ 2 pc of the centre is probably pervaded by gas at 3×10^6 K with density ~ 100 cm^{-3}. This gas, with cooling time $\sim 3 \times 10^4$ yr could, if there were indeed a central hole of $\sim 10^6\ M_\odot$, be heated by the ejecta from disrupted stars[40]. The stellar debris (enriched in helium) would contribute to the energy budget within the central 2 pc, if not to the mass supply. The mass of fast-moving ejecta would be small, but, if its density were high, it might be relatively conspicuous. It would be interesting to consider its possible relevance to the observed fast-moving He feature.

Moreover, the peculiar arm-like features revealed by the Ne line and a thermal radio continuum[11,26] consist of denser gas with temperature 10^3-10^4 K. The cooling of arm-like swaths of gas could have been triggered by the rapid directed outflow from the most recent stellar disruption events (which might have occurred $<10^4$ yr ago). If this interpretation (discussed further in refs 27 and 40) were correct, these features could be 'vapour trails' delineating the tracks of fast ejecta from individual disrupted stars. The cool gas would not itself be ejected material; it would be ambient (perhaps slowly infalling) gas in the cavity whose cooling was triggered by the overpressure. The situation would resemble a 'sector' of a supernova remnant.

Relevance to the nuclei of nearby galaxies

The most obvious consequence of a 10^6-$10^8\ M_\odot$ black hole would be transient flares whenever bound debris from a star was swallowed. The rate is given by equation (2) with $r_{min} = r_T$, the luminosities being as high as $L_E = 10^{44} M_6$ erg s^{-1}. But in a given object, these flares would have a duty cycle of $\sim 10^{-3}$ at peak luminosity (equation(5)). After each luminosity peak, there would be a decline as the 'dregs' were swallowed, but the median luminosity would be far below that which would result from steady efficient accretion of the mass supply implied by (2). Therefore we would not yet expect to have detected such a flare. On the other hand, a sufficiently large sample of such galaxies should reveal some members of the ensemble in a flaring state. Such objects could be searched for out to large distances: they would differ from typical AGNs through the lack of any extended structure (emission line or radio components). If $\sim 10^6\ M_\odot$ holes were prevalent in small or even dwarf galaxies, the nearest such flares, in any given year, may be no further away than the Virgo Cluster.

The mass fraction that is ejected rather than swallowed, though less spectacular than the accretion-powered flares, could

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nevertheless have cumulative effects. There are two ways in which matter can be expelled: (a) When a star is disrupted in a single flyby, about half the debris is ejected in a fast-moving spray of gas. The characteristic velocities are $\sim 10^4\ M_6^{1/6}$ km s^{-1}, the kinetic energy output when a solar-type star is disrupted being $\sim 10^{51}\ M_6^{1/3}$ erg s^{-1}—for $M_6 > 1$ (that is, in all cases of interest here), this exceeds the energy of a supernova. Stars that penetrate well inside r_T would cause still more violent ejection. (b) The bound debris falls back when $t > t_i$ to $r \approx r_T$ where it can acquire centrifugal support. If energy is dissipated at a supercritical rate as ths material swirls closer to the hole, some gas may then be ejected in a radiation-driven wind, or even in double jets aligned with the angular momentum of the debris. In principle it is possible for only a fraction $\varepsilon^{-1}(r_g/r_T)$ to be actually swallowed, ε being the efficiency, all the remainder being ejected. The characteristic speeds here—of order the escape velocity from r_T—would be even higher than those of type (a) outflow discussed above. The debris from a star that does not approach close enough to be destroyed in a single flyby but is eventually disrupted by tidal dissipation could behave similarly.

If there were indeed a $\sim 10^6\ M_\odot$ hole in our Galactic Centre, the consequences of the unbound (outgoing) debris may be conspicuous—indeed, I conjectured above (and in refs 27 and 40) that they may have been unknowingly observed already. The direct effects of unbound debris from an individual disruption would not be readily discernable, however, even in a nearby galaxy such as M31 or M32. The fast-moving ejecta from successive disruptions could nevertheless have an important cumulative influence on the energy balance within the central 10–100 parsecs of a galaxy—more, quite probably, than supernovae exploding in that same volume. The ejecta would be braked, and their kinetic energy thermalized, as they ran into the diffuse gas originating from stellar mass loss or infall from the body of the galaxy. X-ray observations offer direct evidence that there is hot gas in elliptical galaxies, with pressure $nT = 10^7$ cm^{-3} K in the central kiloparsec[41]. Steady accretion of this gas even at the minimum 'Bondi' rate (neglecting cooling) would yield conspicuous effects unless the radiative efficiency were very low[42,43].

The accretion radius is $1.5 \times 10^{17}\ M_6\ T_7^{-1}$ cm. If accretion proceeded unimpeded, the energy within this radius (taking the foregoing estimate of the ambient pressure) would be $10^{43}\ M_6^{-3}\ T_7^{-3}$ erg: even for M_6 as large as 100, this amounts to less than the energy from a single disrupted star. For $M_6 = 100$, the characteristic inflow timescale would be $1.5.10^4\ T_7^{-3/2}$ yr. Comparison with the stellar capture rate (2) confirms that this timescale is long enough for the ejected debris to be equivalent to a fast wind: its ram pressure and heat input could inhibit the steady accretion that would otherwise be inevitable in any galaxy harbouring a black hole.

Stellar disruption and coalescence may have other implications—for instance, the fast moving ejecta may be relevant to the dynamics of line-emitting regions. But they could definitely have the paradoxical effect of making the average nuclear luminosity even lower than would be estimated if only gas were present, because the fast-moving ejecta could stem the steady accretion flow. Thus the quiescence of a nearby nucleus is not incompatible with the presence within it of a massive black hole of $10^8\ M_\odot$.

This paper has outlined several potentially observable effects that call for detailed stellar-dynamical and hydrodynamic calculations. Precise modelling should also allow for a realistic spread in the stellar population. The stellar disruption phenomenon is relevant to high-resolution observations of nearby galaxies, and to the properties of AGNs in general. In the meantime, it would seem worthwhile to search for thermal flares lasting ≤ 1 yr in the nuclei of ordinary quiescent galaxies.

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